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Data Engineering – Hackathon 1

A structured, time-bound challenge designed to test SQL and business insight skills across teams. Below is the complete schedule for the hackathon session.

Activities	Minutes
Team Formation & Hackathon Introduction	30
Hackathon 1.1 (Mandatory) - Hackathon 1.2 (Optional)	90
Break	15
Solution Demo by students and Explanation by Mentors	90
Experience share and Q&A	15

Problem Set Overview – Table 1

FIRST SET OF PROBLEMS (MANDATORY FOR ALL TEAMS)

This set consists of SQL Queries and Business Insights problems. These problems are common for all teams.

Category	Simple	Medium	Complex	Total Problems	Max Duration (Per Problem in Minutes)	Items Per Team Member	Total Time Allocation
SQL Queries		11	5	16	15	4	60
Business Insights		6	2	12	15	2	30
Total Problems		**17**	**7**	**24**	—	**6**	**90 mins**

24

Total Problems

Across SQL Queries and Business Insights

90

Minutes Total

Time allocation for Hackathon 1.1

6

Items Per Member

Each team member handles 6 items

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M1. List All Seats in a Specific Screen

SQL QUERY SIMPLE

Problem Statement

Write a query to list all seats for a given `screen_id`. The output should include `seat_id`, `seat_number`, and `screen_name`.

Expected Output

seat_id	seat_number	screen_name
351	A1	C
352	A2	C
353	A3	C
354	A4	C
355	A5	C
356	A6	C
357	A7	C
358	A8	C
359	A9	C

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M2. Count the Number of Reviews for Each Movie

SQL QUERY SIMPLE

Problem Statement

Write a query to count the number of reviews for each movie. The output should include `movie_id`, `title`, and `review_count`.

Expected Output

movie_id	title	review_count
1	Avatar	79
2	Nine	38
3	Hard Luck	63
4	The Officers' Ward	21
5	You Are So Beautiful	78
6	It's Complicated	42
7	Sherlock Holmes	99
8	Crazy Heart	70
9	The Art of War II: Betrayal	50
10	The Last Station	73

M3. Fetch All Shows from 2025

SQL QUERY SIMPLE

Problem Statement

Write a query to list all shows from 2025. The output should include `show_id`, movie title, screen name, and `show_datetime`.

Expected Output

show_id	title	screen_name	show_datetime
12	Under the Bombs	A	2025-03-26 18:52:00
30	The First Day of the Rest of Your Life	A	2025-03-23 10:56:00
31	Dogtooth	A	2025-03-31 14:14:00
47	Road Trip: Beer Pong	A	2025-04-02 08:51:00
57	10 MPH	A	2025-04-09 23:47:00
67	Lady Windermere's Fan	A	2025-03-24 16:30:00
75	High Life	A	2025-03-31 09:46:00
79	Violence at Noon	A	2025-03-23 05:01:00
108	Falling Up	A	2025-03-26 19:45:00
122	Agora	A	2025-04-04 23:58:00
126	The House on Sorority Row	A	2025-03-29 01:16:00

M4. Find the Most Expensive Food Items

SQL QUERY SIMPLE

Problem Statement

Write a query to find the most expensive food items. The output should include `item_id`, `name`, and `rate`.

Expected Output

item_id	name	rate
15	Fiesta Nachos	19.74
20	Value Pizza	19.27
4	Golden Butter Popcorn	19.06
13	Savory Parmesan Popcorn	19.05
5	Spicy Inferno Burger	18.78

M5. Find the Most Frequent Movie Genres

SQL QUERY SIMPLE

Problem Statement

Write a query to find the most frequent movie genres. The output should include `genre` and the count of movies in that genre.

Expected Output

genre	movie_count
Comedy	108
Horror	101
Drama	99
Sci-Fi	98
Action	94

M6. Get the Most Popular Shows

SQL QUERY SIMPLE

Problem Statement

Write a query to determine which shows have the highest number of tickets sold. The output should include `show_id`, `show_datetime`, and `total_tickets_sold`.

Expected Output

show_id	show_datetime	total_tickets_sold
10	2025-02-04 21:55:00	147
15	2025-02-09 11:05:00	147
14	2025-01-15 12:40:00	139

M7. Find the Customers Who Have Made the Most Spending

SQL QUERY MEDIUM

Problem Statement

Write a query to find the customers who have spent the most on movie tickets. The output should include `user_id`, `name`, and `total_spent`.

Expected Output

user_id	name	total_spent
3603	Thomas Jackson	4479.36
1470	Taylor Phillips	4461.41
4413	Jessica Wright	4310.59
4727	Kathleen Reyes	4243.10
3539	Terry Kramer	4225.95

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M8. Find the Most Popular Showtimes

SQL QUERY MEDIUM

Problem Statement

Write a query to identify the showtimes that have the highest number of bookings. The output should include `show_datetime` and `total_bookings`.

Expected Output

show_datetime	total_bookings
2025-02-04 21:55:00	1009
2025-03-20 05:48:00	974
2025-03-09 15:08:00	969

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M9. Calculate the Average Spending Per Customer

SQL QUERY | MEDIUM

Problem Statement

Write a query to list all upcoming shows. The output should include `show_id`, movie title, screen name, and `show_datetime`.

Expected Output

show_id	title	screen_name	show_datetime
12	Under the Bombs	A	2025-03-26 18:52:00
30	The First Day of the Rest of Your Life	A	2025-03-23 10:56:00
31	Dogtooth	A	2025-03-31 14:14:00
47	Road Trip: Beer Pong	A	2025-04-02 08:51:00
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67	Lady Windermere's Fan	A	2025-03-24 16:30:00
75	High Life	A	2025-03-31 09:46:00
79	Violence at Noon	A	2025-03-23 05:01:00
108	Falling Up	A	2025-03-26 19:45:00
122	Agora	A	2025-04-04 23:58:00
126	The House on Sorority Row	A	2025-03-29 01:16:00

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M10. Find the Most Popular Food Items Ordered

SQL QUERY MEDIUM

Problem Statement

Write a query to determine which food items are ordered the most. The output should include `item_id`, `name`, and `total_orders`.

Expected Output

item_id	name	total_orders
6	Cheesy Delight Pizza	2341
13	Savory Parmesan Popcorn	2285
17	Sweet & Salty Popcorn	2285
9	Volcano Spicy Pizza	2271
10	Caramel Bliss Popcorn	2256

M11. Identify Customers Who Have Earned More Than 800 Loyalty Points

SQL QUERY

MEDIUM

Problem Statement

Write a query to fetch users who have **earned more than 800 loyalty points.** The output should include user details and their total loyalty points balance.

Expected Output

user_id	name	current_points
36	Jessica Levine	1000
522	John Alvarez	1000
1492	Susan Summers	1000
2319	Meagan Martin	1000
38	Jessica Lopez	999
52	Danielle Fernandez	999
201	Dean Williams	999
505	Wendy Murray	999
1594	Alexandra Casey	999
2204	Louis Hall	999

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Complex SQL Queries – Overview

COMPLEX CATEGORY

The complex SQL problems require multi-table joins, subqueries, window functions, and advanced aggregations. These problems are designed to challenge participants with real-world cinema database analytics scenarios.

1

C1 – Revenue by Screen

Analyze total revenue and tickets sold per screen using aggregation and JOIN across bookings and screens.

2

C2 – Revenue by Movie & Time Slot

Calculate total revenue per movie and time slot, requiring multi-level grouping and date functions.

3

C3 – Top Booking Users

Identify users with the most bookings using COUNT aggregation and ranking techniques.

4

C4 – Profitable Food Items

Find the most profitable food items by combining total sales revenue and order counts.

5

C5 – Loyalty Points by Period

Calculate total loyalty points earned per user in the last month using date-range filtering.

C1. Analyze Revenue Contribution by Each Screen

SQL QUERY COMPLEX

Problem Statement

Write a query to calculate the total revenue generated from ticket sales for each screen. The output should include `screen_id`, `screen_name`, `total_revenue`, and `total_tickets_sold`.

Expected Output

screen_id	screen_name	total_revenue	total_tickets_sold
2	B	459610.99	836
3	C	319182.97	593
1	A	130773.12	243
4	D	101719.11	177

C2. Calculate Total Revenue by Movie and Time Slot

SQL QUERY | COMPLEX

Problem Statement

Write a query to calculate the total revenue generated from each movie and time slot. The output should include `movie_id`, `title`, `show_datetime`, and `total_revenue`.

Expected Output

movie_id	title	show_datetime	total_revenue
314	Rockaway	2025-02-04 21:55:00	567382.46
131	Quiet Chaos	2025-03-09 15:08:00	541626.35
434	About Elly	2025-01-15 12:40:00	525845.51
415	Restless Blood	2025-03-20 05:48:00	525745.82
5	You Are So Beautiful	2025-02-07 18:44:00	519501.78
318	Normal Adolescent Behavior	2025-03-15 05:40:00	519331.88
331	Under the Bombs	2025-03-26 18:52:00	516625.23
448	Yuki & Nina	2025-03-21 06:57:00	513106.64
335	Killer Bean 2: The Party	2025-01-25 06:38:00	505862.62
28	Sweepers	2025-01-12 11:27:00	505858.21

C3. Identify Users Who Have Booked the Most Shows

SQL QUERY COMPLEX

Problem Statement

Write a query to identify users who have made the most bookings. The output should include `user_id`, `name`, and `total_bookings`.

Expected Output

user_id	name	total_bookings
20	Sharon Flores	5
12	David Mcneil	5
10	Margaret Perez	5
25	Martha Soto	5
11	Dominique Lee MD	5

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C4. Find the Most Profitable Food Items and Their Sales

SQL QUERY | COMPLEX

Problem Statement

Write a query to find the most profitable food items based on the total sales. The output should include `item_id`, `name`, `total_sales`, and `total_orders`.

Expected Output

item_id	name	total_sales	total_orders
6	Cheesy Delight Pizza	75000.02	4701
8	Meat Lover's Pizza	71741.70	4510
13	Savory Parmesan Popcorn	68198.23	4624
9	Volcano Spicy Pizza	66877.19	4482
16	Earthy Herb Popcorn	66684.97	4429

C5. Calculate Total Points Earned by Users in a Given Period

SQL QUERY COMPLEX

Problem Statement

Write a query to calculate the total loyalty points earned by each user in the last month. The output should include `user_id`, `name`, and `total_points_earned`.

Expected Output

user_id	name	total_points_earned
765	Donald Valentine	192
1105	Mr. Ricky Holloway	185
3179	Peggy Dixon	178
4978	Benjamin Harper	176
1942	Stacey Lowe	173
998	Kimberly Harris	160
1251	Shirley Booker	160
2286	Victor Baker	159
1435	Michael Mendez	158
4744	Brian Williams	158
95	Russell Delgado	156

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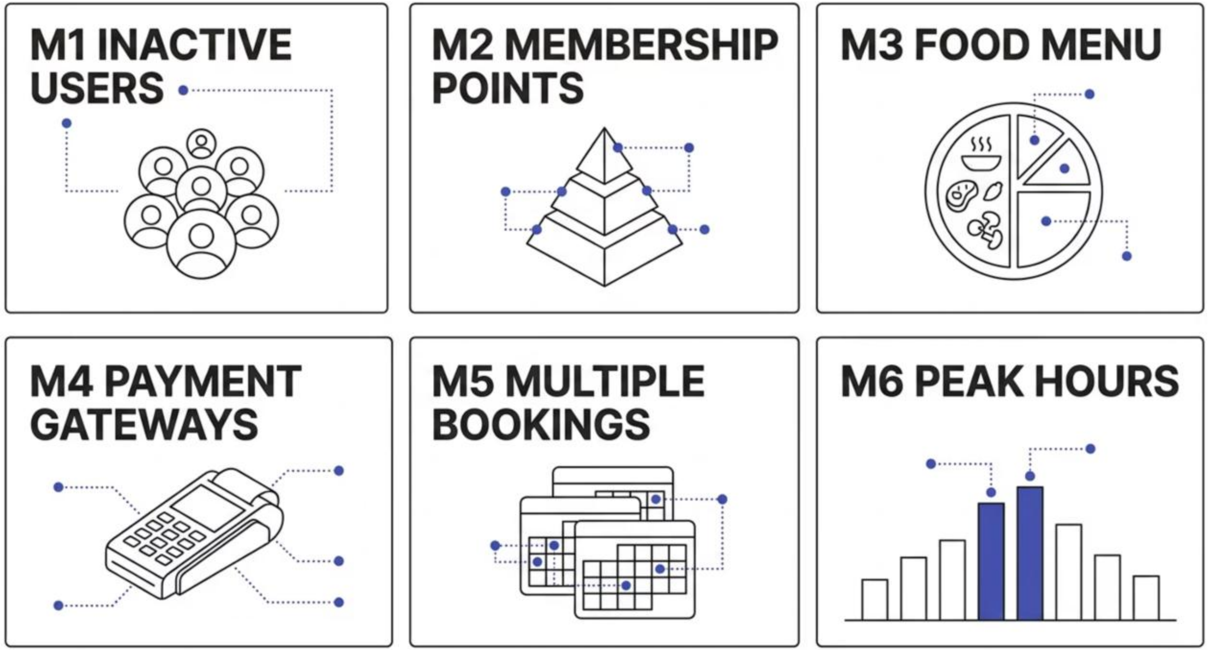
Hackathon 1 – Business Insights

MEDIUM CATEGORY

Business Insights problems go beyond raw SQL — they require participants to interpret query results and articulate the real-world value of the data. Each problem includes a business context and expected analytical takeaways.



M1. Identify Users Who Haven't Made a Booking

BUSINESS INSIGHTS MEDIUM

Problem Statement

"The theater management wants to identify users who have registered but never booked a ticket. Write a query to fetch all users who have no associated bookings."

Business Insights from the Query

- Helps in sending promotional offers to inactive users.
- Allows targeted marketing efforts to increase user engagement.

Expected Output

user_id	name	email
5001	Gregory Morrison	vcampos306@icloud.com
5002	Luis Barajas	nicoleadams601@hotmail.com
5003	Christina Middleton	rogerphelps191@outlook.com
5004	Melissa Neal	smithtaylor863@outlook.com
5005	Haley Peters	isaacstephens169@outlook.com
5006	James Munoz	patricia09979@gmail.com
5007	Matthew Moore	garzaelijah634@gmail.com
5008	Tina Johnson	rebecca25166@hotmail.com
5009	Jose Pham	elliottlaura215@hotmail.com

M2. List Users with Membership and Their Current Points

BUSINESS INSIGHTS MEDIUM

Problem Statement

"The loyalty program manager needs a list of all members and their current loyalty points. Write a query to retrieve the name and current points for all users who have a membership."

Business Insights from the Query

- Get a snapshot of the current membership base and their points.
- Identify members for potential targeted promotions based on their points.
- Track the overall engagement with the loyalty program.

Expected Output

name	current_points
Zachary Adams	809
Shane Thomas	348
Alexa Zhang	143
Hayden Perez	639
Todd Fleming	897
Mitchell Gay	853
Lori Robinson	1
Joseph Cook	665
Paul Baxter	765

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M3. List All Food Items with Their Available Sizes and Rates

BUSINESS INSIGHTS MEDIUM

Problem Statement

"The food and beverage department needs a comprehensive list of all food items along with their available sizes and corresponding rates. Write a query to retrieve this information."

Business Insights from the Query

- Provides a complete menu overview for internal use.
- Helps in managing inventory and pricing.
- Can be used to generate customer-facing menus.

Expected Output

item_name	size_name	rate
BBQ Ranch Burger	Large	13.85
BBQ Ranch Burger	Medium	12.01
BBQ Ranch Burger	Small	7.02
Caramel Bliss Popcorn	Large	9.99
Caramel Bliss Popcorn	Medium	12.51
Caramel Bliss Popcorn	Small	10.90
Caramel Crunch Popcorn	Large	13.91
Caramel Crunch Popcorn	Medium	9.21
Caramel Crunch Popcorn	Small	15.71
Cheddar Cheese Popcorn	Large	15.68
Cheddar Cheese Popcorn	Medium	11.34
Cheddar Cheese Popcorn	Small	12.13
Cheesy Delight Pizza	Large	15.32
Cheesy Delight Pizza	Medium	14.18
Cheesy Delight Pizza	Small	18.54
Classic Chicken Burger	Large	6.59

M4. Find the Payment Gateways Used Most Frequently

BUSINESS INSIGHTS MEDIUM

Problem Statement

"The finance department wants to understand which payment gateways are preferred by customers. Write a query to count the number of times each payment gateway has been used."

Business Insights from the Query

- Identify popular payment methods for potential partnerships or feature enhancements.
- Understand customer preferences for payment options.
- Inform decisions about which payment gateways to prioritize and maintain.

Expected Output

gateway_name	usage_count
Razorpay	5039
PayPal	5008
Stripe	4899

M5. List Customers with Multiple Bookings on the Same Day

BUSINESS INSIGHTS MEDIUM

Problem Statement

"The theater management wants to identify customers who have made multiple bookings on the same day. Write a query to find users who have booked more than once on any given day."

Business Insights from the Query

- Recognize highly engaged customers.
- Offer special promotions or loyalty rewards to frequent bookers.
- Understand customer behavior patterns for better service.

Expected Output

User ID	Name	Booking Date	Number of Bookings
55	Kathryn Cabrera	2025-02-10	2
441	Autumn Gross	2025-02-10	2
657	John Lee	2025-02-10	2
1887	Christopher Rogers	2025-02-10	2
4614	Whitney Blankenship	2025-02-10	2
4650	Mario Robertson	2025-02-10	2
513	Joseph Murray	2025-02-11	2
617	Christopher Johnson	2025-02-11	2
723	Andrew Dean	2025-02-11	2
898	Mr. Zachary Marshall	2025-02-11	2
982	Jonathan Reyes	2025-02-11	2
1430	Barbara Mcdonald	2025-02-11	2
1993	Kyle Wagner	2025-02-11	2
2136	Monica Wilkerson MD	2025-02-11	2
2253	Melanie Oconnor	2025-02-11	2
2423	Victoria Martinez	2025-02-11	2

M6. Identify Peak Booking Hours

BUSINESS INSIGHTS MEDIUM

"The management wants to understand customer booking behavior by identifying peak hours for ticket bookings. Write a query to find the hours with the highest ticket bookings."

Business Insights from the Query

- Helps in staffing and operational planning during peak hours.
- Can be used for targeted promotions during off-peak hours.

Expected Output

hour	total_bookings
3	672
19	669
4	668
8	658
0	644
9	636
5	635
17	634
22	632
13	631
1	626
12	623
23	620

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C1. Analyze the Trend of New User Registrations Over Time

BUSINESS INSIGHTS

COMPLEX

"The marketing team wants to track the growth of the user base. Write a query to analyze the number of new user registrations per month over the last year."

Business Insights from the Query

- Identify trends in user acquisition and growth.
- Evaluate the effectiveness of marketing campaigns over time.
- Forecast future user growth based on historical data.

Expected Output

registration_month	new_users
2025-02	4375
2025-03	625

C2. Identify Customers with the Highest Spending on Food Items

BUSINESS INSIGHTS | COMPLEX

"The food and beverage department wants to recognize customers who contribute the most to food sales. Write a query to find the top 10 customers based on their total spending on food items."

Business Insights from the Query

- Identify high-value food and beverage customers.
- Develop targeted promotions or loyalty rewards for these customers.
- Understand which customers are most likely to purchase food items.

Expected Output

user_id	name	total_food_spending
3452	Mr. Thomas Lopez DVM	837.34
2575	Eric Gonzalez	808.26
4046	Melissa Watson	791.01
2705	Dr. Kyle Mclaughlin	774.82
4355	Laura Burke	774.75
4515	Kelly Baldwin	767.37
3923	Gabriel Clark	764.68
4658	Rodney Brown	759.52
1393	Melanie Cruz	755.02
1435	Michael Mendez	749.28

Thank You

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